



Huawei Cloud EI – OCR API Services



Foreword

- This chapter mainly describes the functional features, application scenarios, and corresponding practical cases of Huawei Cloud's OCR API service.



Objectives

- After completing this course, you will be able to:
 - Understand the main functions of the OCR API service
 - Distinguish the application scenarios of the OCR API service
 - Understand practical cases of the OCR API service
 - Understand the calling process of the OCR API service



Contents

1. Overview of OCR

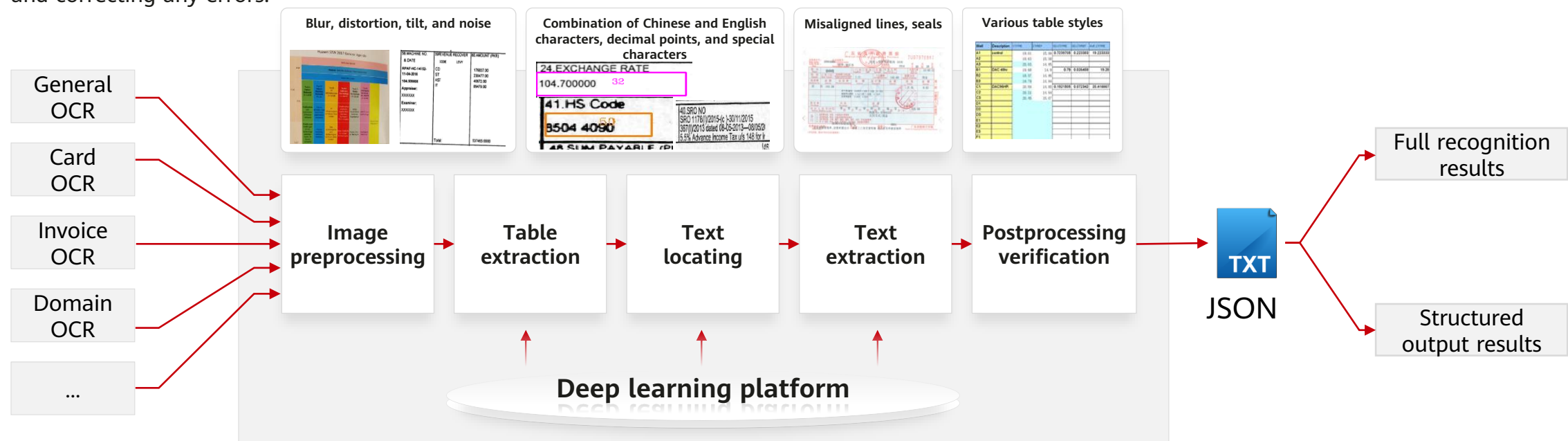
2. Application Scenarios of Text RecognitionService Invocation

3. The process of calling the text recognition service

Introduction to OCR

Optical Character Recognition (OCR)

OCR technology automatically detects and converts text in images or scanned documents into editable text, eliminating the need for manual data entry. It has a wide range of applications, including card identification, license plate recognition, healthcare, PDF-to-Word conversion, photo and screenshot recognition, web image recognition, autonomous driving, paperless offices, document editing and proofreading, logistics sorting, public opinion monitoring, document retrieval, subtitle recognition, and literature search. OCR can recognize both printed and handwritten characters using a general process that involves identifying the character area, dividing it into individual characters, classifying and recognizing the characters, and post-processing and correcting any errors.



Training with large volumes of raw/analog data

<http://3ms.huawei.com/km/blogs/details/7839677?l=en>

Functions of Huawei Cloud OCR

Full-text recognition

General OCR

General Text

General Table

Web Image

Extensible

Structured product

Card OCR

ID Card

Driving/Vehicle
License

Passport

Bank Card

Business Card

Extensible

Domain OCR

Electronic Waybill

Financial Report

Extensible

1. OCR services developed in China can be exported and customized for use in other countries using customer-provided samples.
2. Huawei Cloud can also provide OCR software that supports local languages through partnerships with local companies.

Myanmar ID Card

Myanmar Driving
License

Thailand ID Card

Passport

India PAN Card/ID Card

Available

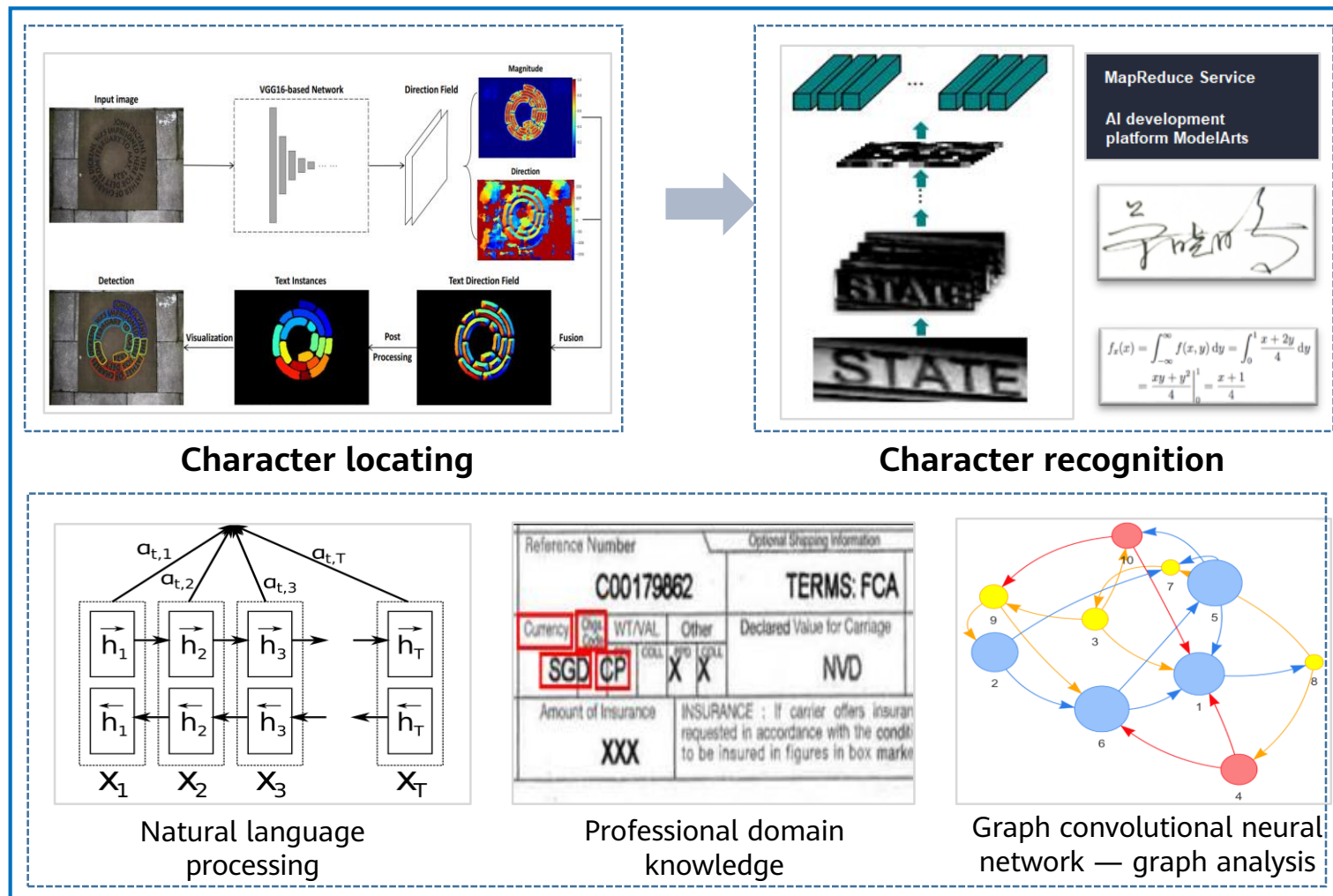
OCR Service Process



Structured information input

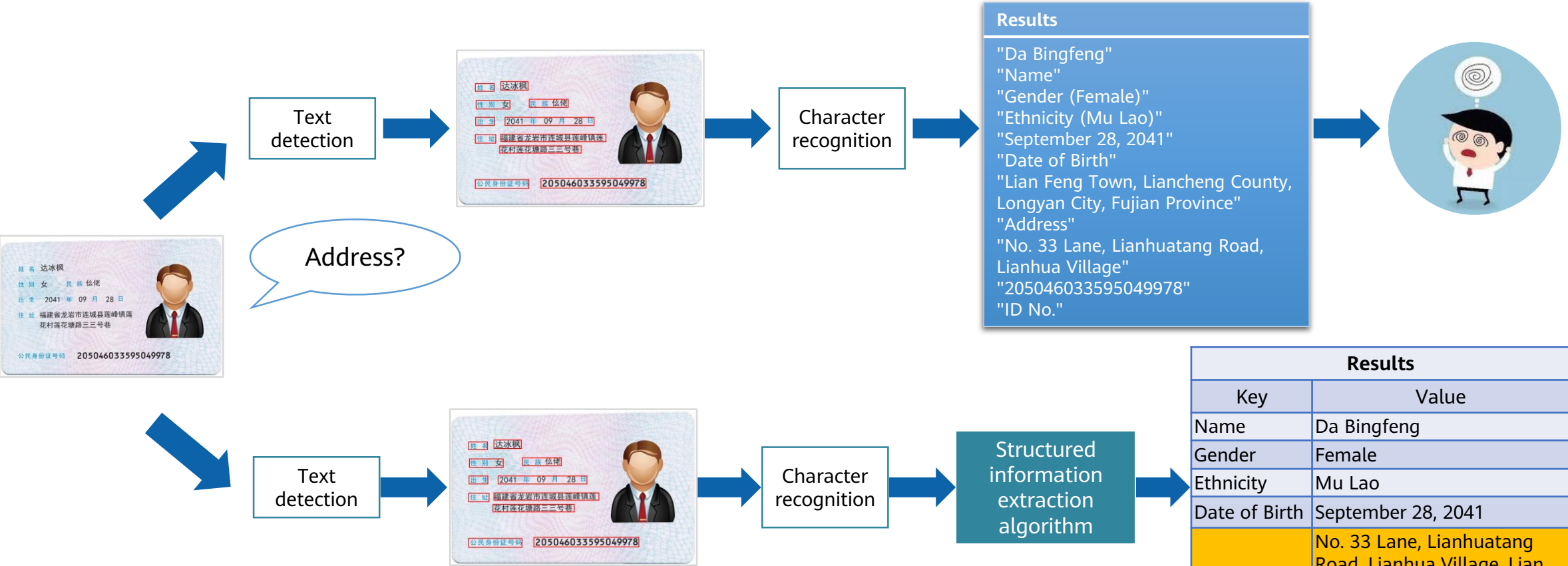


Structured result output



Structured information extraction

What Is Structured Information Extraction



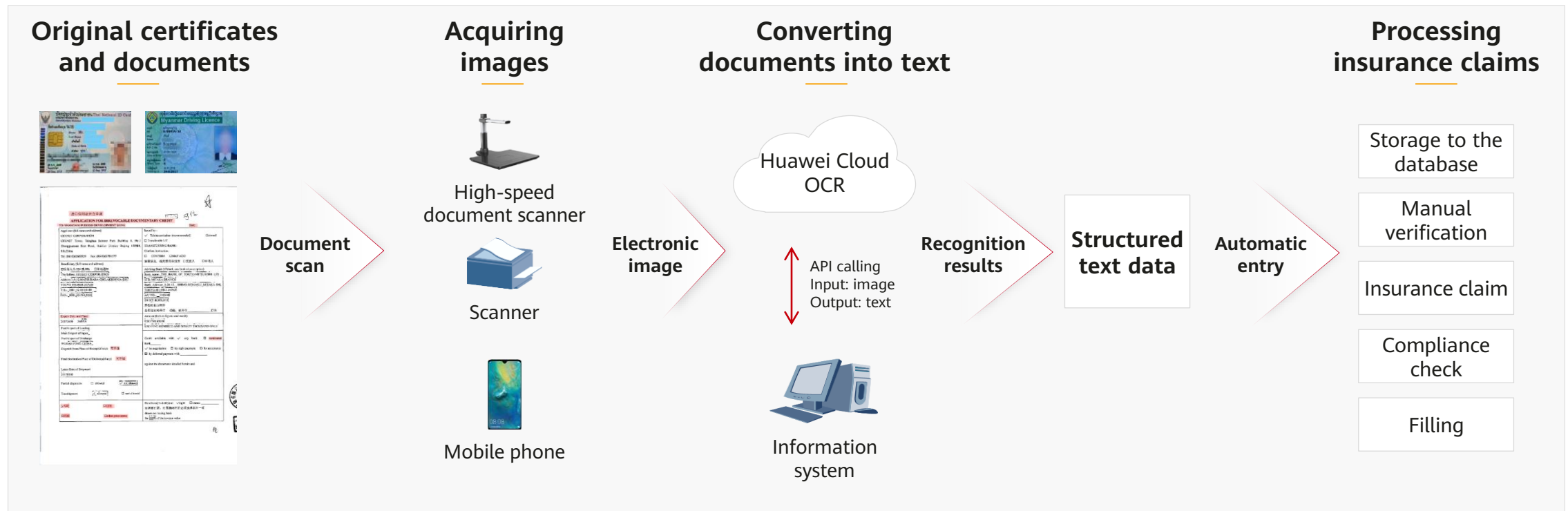
Structured information extraction:
The recognition results are returned as key-value pairs.

Results	
Key	Value
Name	Da Bingfeng
Gender	Female
Ethnicity	Mu Lao
Date of Birth	September 28, 2041
Address	No. 33 Lane, Lianhuatang Road, Lianhua Village, Lian Feng Town, Liancheng County, Longyan City, Fujian Province
ID No.	205046033595049978

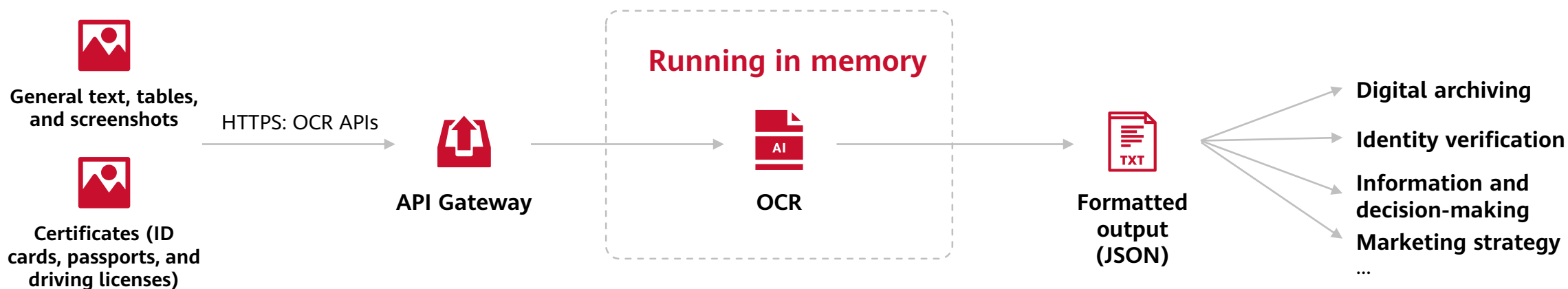
Introduction to the OCR Solution

Higher efficiency at lower costs

1. Many traditional industries, such as carriers, banks, and retail, have millions of scanned documents that need to be processed electronically for archiving. However, manual data entry is time-consuming and inefficient.
2. Real-time registration and review processes often involve multiple items that require manual entry, which can be costly and lead to a poor user experience.
3. In certain countries, ID cards may have handwritten information and poor-quality images, making them difficult to identify.



Introduction to the OCR Lite Solution



✓ High accuracy

Our advanced deep learning algorithms optimize various service scenarios to achieve highly accurate character recognition.

✓ Ease of use

Our RESTful APIs are highly compatible and user-friendly.

✓ Support for complex scenarios

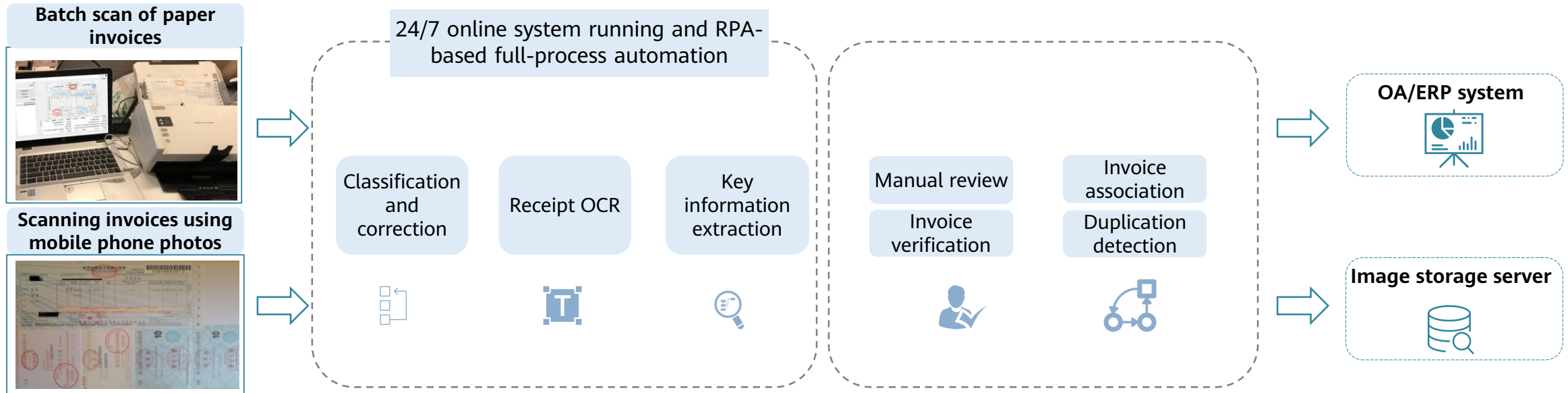
Our OCR technology can recognize certificates with complex backgrounds, distortions, and tilts, as well as forms with seals and misaligned text.

✓ Data security

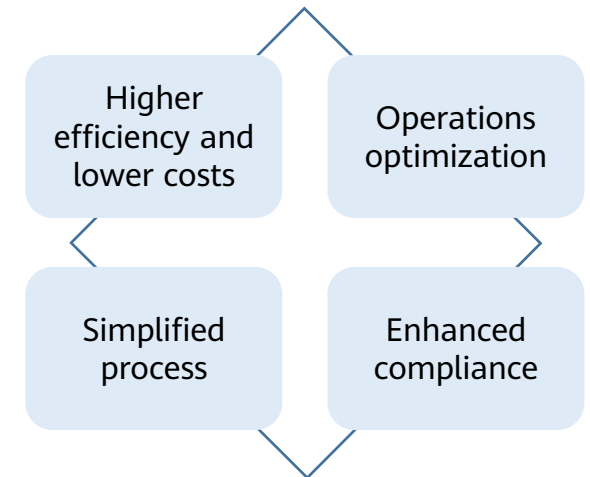
We only use customer data for identification purposes.

API call guidelines: https://support.huaweicloud.com/intl/en-us/api-ocr/ocr_03_0050.html

Use Case — Invoice OCR



- **Multiple access modes:** Connects to scanners, high-speed document scanners, or mobile phones for batch image acquisition.
- **Invoice variety:** Supports various invoice types, including VAT general/special/electronic invoices, ETC invoices, roll invoices, taxi invoices, train tickets, flight itineraries, quota invoices, and toll invoices.
- **Multiple invoices in one image:** Automatically identifies and classifies different invoice types that are combined in a single image.
- **Visual comparison:** Converts location data into an Excel file for easy statistical analysis.



Use Case — Smart Logistics OCR

➤ ID Card OCR

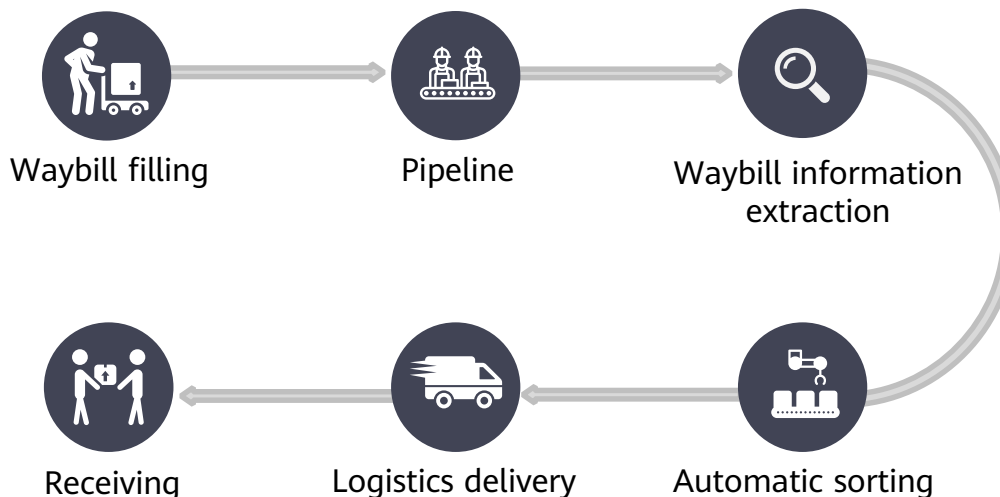
- ID card photographing, recognition, and verification with a mobile app



➤ Screenshot OCR

- Self-service parcel posting
- E-commerce sellers use this service to identify buyers' screenshots and automatically retrieve information such as the name and address.

24/7, less than 2s per waybill recognition



➤ E-waybill OCR

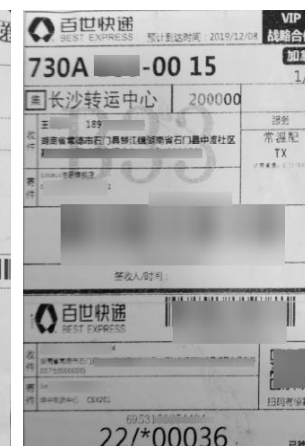
- Automatically extract main information such as the waybill number, recipient/sender name, phone number, and address.
- Supports electronic waybills of mainstream courier companies.
- Recognizes multiple waybills concurrently.



Up to 98% in accuracy, reducing the reshooting workload and eliminating external interference



Streamlined automation process, minimizing human intervention to reduce costs



Fully automated process, protecting data privacy

Practice Case — Medical Insurance Claim Settlement

Original documents

Acquiring images

Converting documents into text

Processing insurance claims



ID card and bank card



Medical invoice



Other invoices

Document scan

High-speed document scanner
Scanner
Mobile phone
...

Electronic image



Information system

Huawei Cloud OCR

API calling
Input: image
Output: text

Recognition results

Structured text data

Automatic entry

Storage to the database

Manual verification

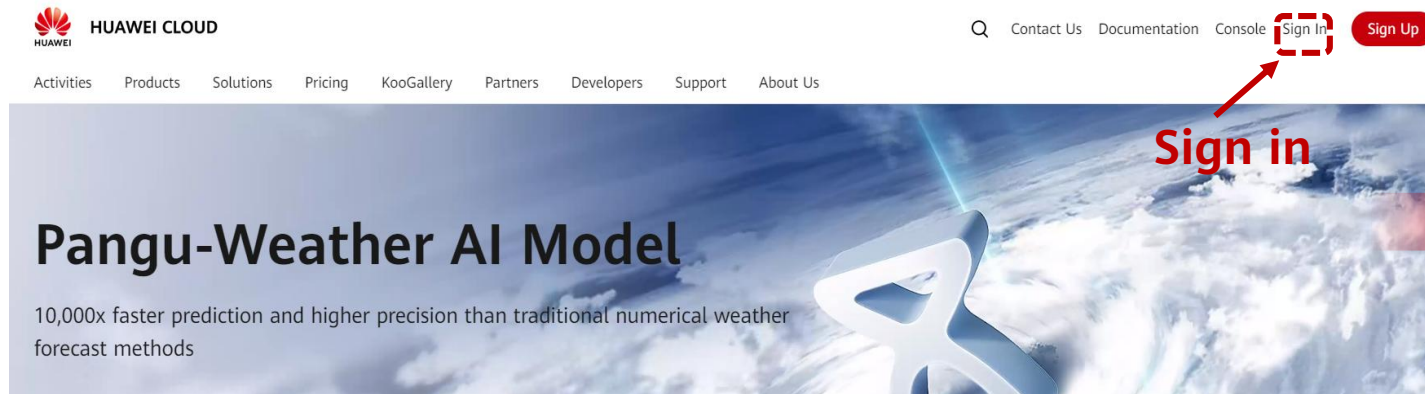
Insurance claim

Compliance check

Filling

API Calling Method

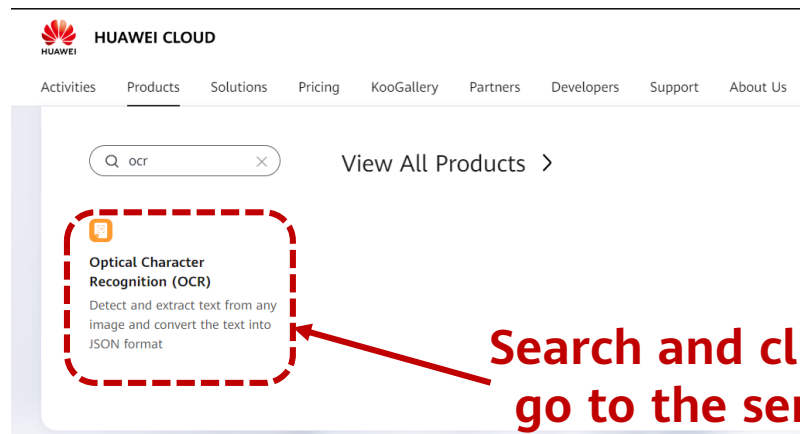
Huawei Cloud official website
<https://www.huaweicloud.com/intl/en-us/>



HUAWEI ID login

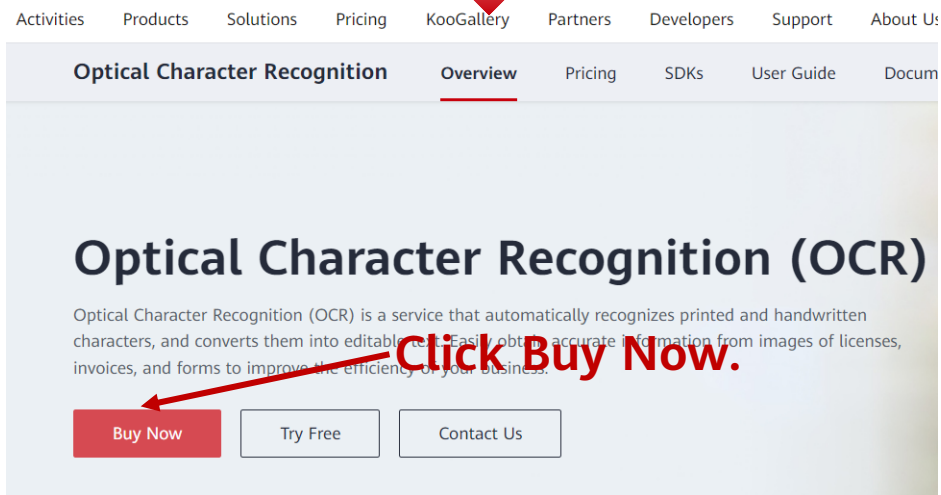
The screenshot shows the HUAWEI ID login page. It has a clean, modern design with a white background. The top section contains a large input field for 'Phone/Email/Login ID/HUAWEI CLOUD account name' and a 'Password' field with a toggle for visibility. Below these fields is a prominent red 'LOG IN' button. Under the button, there are links for 'Register' and 'Forgot password?'. A section titled 'Use Another Account' follows, with a red dashed box highlighting the 'IAM User' option. Other options listed are 'Federated User', 'Huawei Website Account', 'Huawei Enterprise Partner', and 'Huawei Cloud Account'. At the bottom, a small disclaimer states: 'Your account and network information will be used to help improve your login experience. [Learn more](#)'.

API Calling Method (2) — Using OCR as an Example

[Sign In](#)[Subscribe](#)[Call](#)

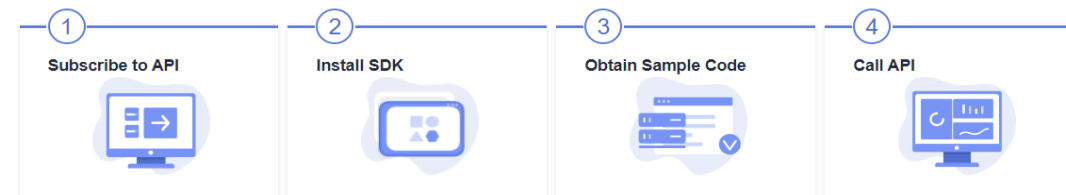
Search and click to go to the service details page.

1



Click Buy Now.

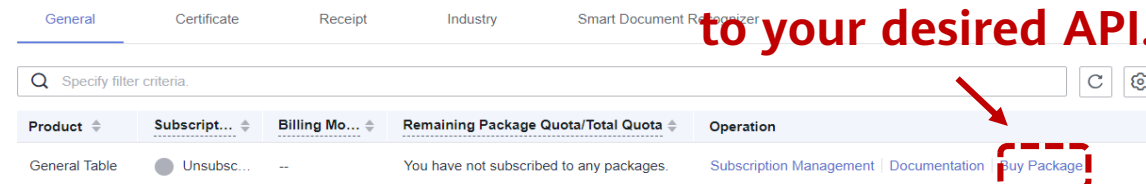
Operation Instructions



Usage Instructions

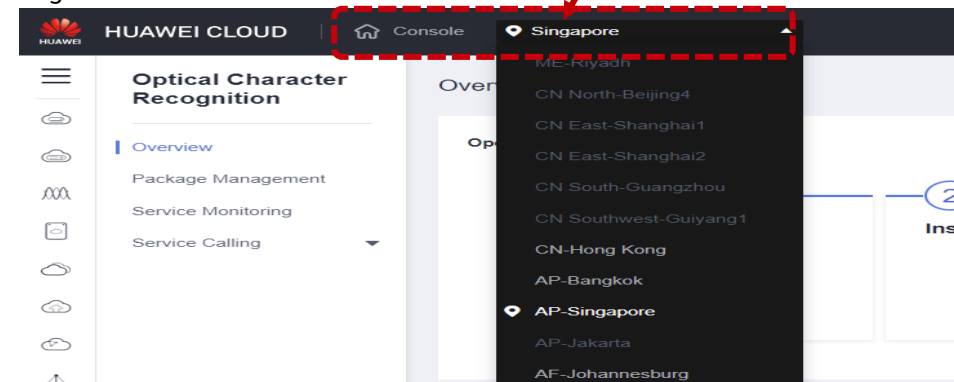
1. Check the region where the console is. If there is no required package available in the region, switch to another region.
2. The package for one region cannot be used in another region. For example, the package for AP-Bangkok can only be used in AP-Bangkok. Exercise caution when purchasing a package for a region.
3. If you want to use data stored on OBS, you can [authorize](#) OCR to access it.
4. Fees are deducted in the following sequence: specified enterprise project package > package with scenario > package with region > package with service > package with product.

Select and subscribe to your desired API.



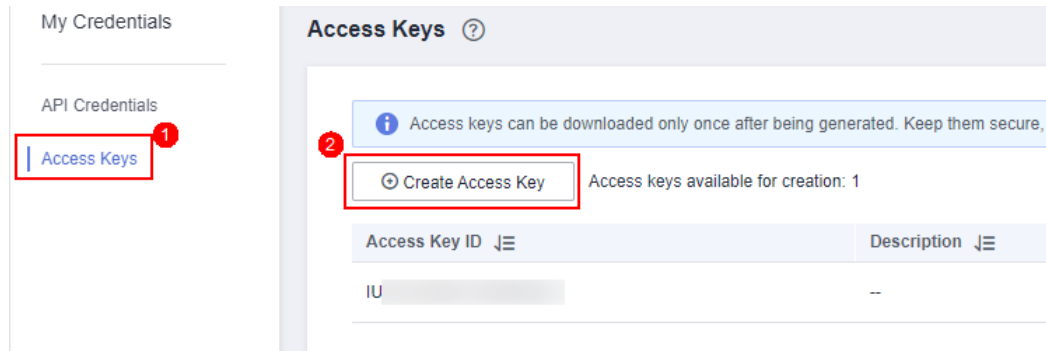
1. Check the region where the console is. If there is no required package available in the region, switch to another region.
2. The package for one region cannot be used in another region. For example, the package for AP-Bangkok can only be used in AP-Bangkok. Exercise caution when purchasing a package for a region.

3



API Calling Method (3) — Using OCR as an Example

Obtain an AK and SK on the **My Credentials** > **Access Keys** page.



A	B	C	D	E
User Name	Access Key	Secret Access Key		
testuser	LSKM	CrIzaQ		
	AK	SK		

Install the SDK.

```
# Install the huaweicloudsdkcore package.

pip # I
pip Structure

Terminal: Local x + v
PS D:\demo> pip install huaweicloudsdkcore
Collecting huaweicloudsdkcore
  Downloading huaweicloudsdkcore-3.0.70-py2.py3-none-any.whl (42 kB)
    | 42 kB 123 kB/s
Collecting six>=1.16.0
  Downloading six-1.16.0-py2.py3-none-any.whl (11 kB)
```

2 Run the sample code to obtain recognition results.

```
from huaweicloudsdkcore.auth.credentials import BasicCredentials
from huaweicloudsdkcore.v1.region_ocr_region import OCRRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdkcore.v1 import *

if __name__ == '__main__':
    ak = 'LSKM'
    sk = 'CrIzaQ'
    credentials = BasicCredentials(ak, sk)

    client = _ClientBuilder() \
        .with_credentials(credentials) \
        .with_region(OCRRegion.value_of('cn-southwest-1')) \
        .build()

    try:
        request = RecognizePassportRequest()
        request.body = PassportRequestBody()
        request.body.set_image('data://...')
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)

    response = client.recognize_passport(request)
    print(response)
except exceptions.ClientRequestException as e:
    print(e.status_code)
    print(e.request_id)
    print(e.error_code)
    print(e.error_msg)

if __name__ == '__main__':
    pass
```

```
{
  "result": {
    "country_code": "CHN",
    "passport_number": "P00000000",
    "surname": "CHEN",
    "given_name": "WEN",
    "sex": "F",
    "date_of_birth": "1990-01-01",
    "date_of_expiry": "2020-01-01",
    "confidence": 0.9999,
    "country_code": "CHN",
    "surname": "CHEN",
    "given_name": "WEN",
    "sex": "F",
    "date_of_birth": "1990-01-01",
    "date_of_expiry": "2020-01-01",
    "confidence": 0.9999,
    "extra_info": {}
  }
}
```

Huawei Cloud API Explorer

API Explorer: an online debugging tool for retrieving and debugging APIs and generating example code
<https://console-intl.huaweicloud.com/apiexplorer/#/openapi>

[illegible]

Q&A

1. Which of the following API services are provided by Huawei Cloud? ()
- A. Speech Interaction Service (SIS)
 - B. Content Moderation
 - C. Optical Character Recognition (OCR)
 - D. ImageSearch



2. What are the application scenarios of Huawei Cloud Image Recognition?
- A. Scenario analysis
 - B. Image tagging
 - C. Object detection
 - D. Smart album

Q&A

1. What are the benefits of using an OCR-based system for financial reimbursements? ()
 - A. Multiple tickets or certificates in a single image
 - B. Support for automatic segmentation
 - C. Multiple access modes
 - D. Support for integration with an OA system and a storage server



总结

- This chapter first introduces the main features and common application scenarios of Huawei Cloud's OCR API service. It then presents practical implementation cases related to the OCR API service. Finally, it demonstrates the process for users to activate the OCR API service on Huawei Cloud and call the API locally. ◦



Acronyms and Abbreviations

- EI: enterprise intelligence
- API: application programming interface
- GCN: graph convolutional network
- IM: instant message
- RPA: robotic process automation
- OCR: optical character recognition



Recommendations

- OCR documentation:
 - <https://support.huaweicloud.com/intl/en-us/ocr/index.html>



Thank You.

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